

■ 시나리오 별 백업 복구(User Managed) - Archive log mode

KEY WORDS

- <시나리오 1> Archive log mode 의 변경
- <시나리오 2> 데이터베이스 운영 중에 offline 되는 데이터파일 손상되었을때 복구
- <시나리오 3> 오프라인시 테이블스페이스가 손상되었을 경우
- <시나리오 4> 데이터베이스가 정상적인 종료 후 여러 데이터파일이 손상되었을 경우

<<시나리오 1>> Archive log mode의 변경

SYS@orcl> archive log list

```
Database log mode          No Archive Mode
Automatic archival         Disabled
Archive destination        /u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch
Oldest online log sequence 22
Current log sequence       24
```

SYS@orcl> select name, log_mode from v\$database;

```
NAME      LOG_MODE
-----
ORCL      NOARCHIVELOG
```

SYS@orcl> show parameter DB_RECOVERY_FILE_DEST

```
NAME                                TYPE      VALUE
-----
db_recovery_file_dest               string
db_recovery_file_dest_size          big integer 0
```

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#, b.next_change# from v\$logfile a, v\$log b where a.group# = b.group# order by 1,2;

	GROUP#	SEQUENCE#	MEMBER	MB	
	STATUS	ARC	FIRST_CHANGE#	NEXT_CHANGE#	
		1	22 /u01/app/oracle/oradata/ORCL/redo01.log	10	INACTIVE
NO	2338180	2339136			
		2	24 /u01/app/oracle/oradata/ORCL/redo02.log	10	CURRENT
NO	2339146	1.8447E+19			
		3	23 /u01/app/oracle/oradata/ORCL/redo03.log	10	INACTIVE
NO	2339136	2339146			

```
SYS@orcl> !mkdir -p /home/oracle/arch1 /home/oracle/arch2
```

```
SYS@orcl> !ls /home/oracle/arch$
```

```
/home/oracle/arch1:
```

```
/home/oracle/arch2:
```

```
SYS@orcl> alter system set log_archive_dest_1 = "location=/home/oracle/arch1 manatory" scope=spfile;
System altered.
```

```
SYS@orcl> alter system set log_archive_dest_2 = "location=/home/oracle/arch2 optional" scope=spfile;
System altered.
```

```
SYS@orcl> show parameter log_archive_format
```

NAME	TYPE	VALUE
log_archive_format	string	%t_%s_%r.dbf

```
SYS@orcl> select * from v$database_incarnation;
```

INCARNATION#	RESETLOGS_CHANGE#	RESETLOGS_TIME	PRIOR_RESETLOGS_CHANGE#	PRIOR_RESETLOGS_TI	STATUS	RESETLOGS_ID	PRIOR_INCARNATION#
1	1	17-APR-19	0		PARENT		
1005785759	0						
NO		0					
2	1920977	01-MAY-25	1	17-APR-19	PARENT		
1199933387	1						
NO		0					
3	2310250	07-MAY-25	1920977	01-MAY-25	CURRENT		
1200525254	2						
NO		0					

```
SYS@orcl> alter system set log_archive_format = '%T_%S_%R.arc' scope=spfile;
```

System altered.

```
SYS@orcl> ! ps -ef | grep arc
```

```
oracle    7347 21215  0 00:38 pts/0    00:00:00 /bin/bash -c  ps -ef | grep arc
oracle    7349  7347  0 00:38 pts/0    00:00:00 grep arc
```

```
SYS@orcl> shutdown immediate
```

Database closed.

Database dismounted.

ORACLE instance shut down.

```
SYS@orcl> startup mount
```

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 754974720 bytes

Database Buffers 369098752 bytes

Redo Buffers 7880704 bytes

Database mounted.

```
SYS@orcl> select status from v$instance;
```

STATUS

MOUNTED

```
SYS@orcl> alter database archivelog;
```

Database altered.

```
SYS@orcl> alter database open;
```

Database altered.

```
SYS@orcl> select name, log_mode from v$database;
```

NAME LOG_MODE

ORCL ARCHIVELOG

```
SYS@orcl> archive log list
```

Database log mode Archive Mode

```

Automatic archival          Enabled
Archive destination         /home/oracle/arch2
Oldest online log sequence  22
Next log sequence to archive 24
Current log sequence        24
SYS@orcl> show parameter log_archive_max_process

```

NAME	TYPE	VALUE
log_archive_max_processes	integer	4

```

SYS@orcl> ! ps -ef | grep arc

```

```

oracle  7507    1  0 00:40 ?        00:00:00 ora_arc0_orcl
oracle  7512    1  0 00:40 ?        00:00:00 ora_arc1_orcl
oracle  7514    1  0 00:40 ?        00:00:00 ora_arc2_orcl
oracle  7516    1  0 00:40 ?        00:00:00 ora_arc3_orcl
oracle  7914 21215 0 00:43 pts/0    00:00:00 /bin/bash -c ps -ef | grep arc
oracle  7916  7914 0 00:43 pts/0    00:00:00 grep arc

```

```

SYS@orcl> select * from v$log;

```

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC	STATUS
FIRST_CHANGE#	FIRST_TIME	NEXT_CHANGE#					
1	1	25	10485760	512	1	NO	CURRENT
09-MAY-25		1.8447E+19					
		0					
2	1	24	10485760	512	1	YES	INACTIVE
09-MAY-25		2341425					
09-MAY-25		0					
3	1	23	10485760	512	1	YES	INACTIVE
09-MAY-25		2339146					
09-MAY-25		0					

```

SYS@orcl> alter system switch logfile;

```

System altered.

SYS@orcl> /

System altered.

SYS@orcl> !! ls /home/oracle/arch1 /home/oracle/arch2
/home/oracle/arch1:

/home/oracle/arch2:

arch_1_24_1200525254.arc arch_1_25_1200525254.arc arch_1_26_1200525254.arc

SYS@orcl> select * from v\$archived_log;

SYS@orcl> select * from v\$log;

SYS@orcl> alter system archive log current;

System altered.

SYS@orcl> show parameter log_archive_max_process

NAME	TYPE	VALUE

log_archive_max_processes	integer	4

SYS@orcl> select file#, name, checkpoint_change#, status from v\$datafile;

■ 백업 받기

SYS@orcl > alter database begin backup;

Database altered.

SYS@orcl> !cp /u01/app/oracle/oradata/ORCL/* /home/oracle/backup/open_bkp/

SYS@orcl> alter database end backup;

Database altered.

SYS@orcl> alter database backup controlfile to '/home/oracle/backup/open_bkp/control_20250509.ctl';

```
SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd  
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

```
SYS@orcl> ! ls /home/oracle/arch*
```

<<시나리오 2>> 데이터베이스 운영 중에 offline 되는 데이터파일 손상되었을때 복구

```
SQL> alter system archive log current;
System altered.
```

-- 사전 확인

```
SQL> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

```

FILE#
-----
NAME
-----
CHECKPOINT_CHANGE# STATUS          CHANGE# TIME
-----
1
/u01/app/oracle/oradata/ORCL/system01.dbf
2426581 NOT ACTIVE                2424496 2025-05-09 15:24:18.55458

3
/u01/app/oracle/oradata/ORCL/sysaux01.dbf
2426581 NOT ACTIVE                2424496 2025-05-09 15:24:18.55458
```

```

FILE#
-----
NAME
-----
CHECKPOINT_CHANGE# STATUS          CHANGE# TIME
-----
4
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
2426581 NOT ACTIVE                2424496 2025-05-09 15:24:18.55458

7
/u01/app/oracle/oradata/ORCL/users01.dbf
```

```

FILE#
-----
NAME
-----
```

CHECKPOINT_CHANGE#	STATUS	CHANGE#	TIME
2426581	NOT ACTIVE	2424496	2025-05-09 15:24:18.55458

SQL> select * from v\$log;

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC
1	1	19	10485760	512	1	YES
INACTIVE		2425407	09-MAY-25		2426581	
09-MAY-25		0				
2	1	20	10485760	512	1	YES
INACTIVE		2426581	09-MAY-25		2428588	
09-MAY-25		0				

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC
3	1	21	10485760	512	1	NO
CURRENT		2428588	09-MAY-25		1.8447E+19	
		0				
3	1	18	10485760	512	1	YES
INACTIVE		2425311	09-MAY-25		2425407	
09-MAY-25		0				

SQL> select recid, sequence#, resetlogs_change#, FIRST_CHANGE#, NEXT_CHANGE#, STATUS from v\$archived_log;

RECID	SEQUENCE#	RESETLOGS_CHANGE#	FIRST_CHANGE#	NEXT_CHANGE#	S
1	6	2396537	2418165	2422033	A
2	6	2396537	2418165	2422033	A

26	18	2396537	2425311	2425407 A
27	19	2396537	2425407	2426581 A
28	19	2396537	2425407	2426581 A

28 rows selected.

- 아카이브 파일이 물리적으로 있는지 꼭 체크!!!!

SQL> ! ls /home/oracle/arch*

/home/oracle/arch1:

```
0001_0000000010_1200585133.arc      0001_0000000013_1200585133.arc      0001_0000000016_1200585133.arc
0001_0000000019_1200585133.arc
0001_0000000011_1200585133.arc      0001_0000000014_1200585133.arc      0001_0000000017_1200585133.arc
0001_0000000020_1200585133.arc
0001_0000000012_1200585133.arc 0001_0000000015_1200585133.arc 0001_0000000018_1200585133.arc
```

/home/oracle/arch2:

```
0001_0000000010_1200585133.arc      0001_0000000013_1200585133.arc      0001_0000000016_1200585133.arc
0001_0000000019_1200585133.arc
0001_0000000011_1200585133.arc      0001_0000000014_1200585133.arc      0001_0000000017_1200585133.arc
0001_0000000020_1200585133.arc
0001_0000000012_1200585133.arc 0001_0000000015_1200585133.arc 0001_0000000018_1200585133.arc
```

/home/oracle/arch2:

```
0001_0000000010_1200585133.arc      0001_0000000013_1200585133.arc      0001_0000000016_1200585133.arc
0001_0000000019_1200585133.arc
0001_0000000011_1200585133.arc 0001_0000000014_1200585133.arc 0001_0000000017_1200585133.arc
0001_0000000012_1200585133.arc 0001_0000000015_1200585133.arc 0001_0000000018_1200585133.arc
```

SQL> create table hr.emp7(id number) tablespace users;

Table created.

SQL> insert into hr.emp7 values(100);

1 row created.

SQL> commit;

Commit complete.

```
SQL> select * from hr.emp7;
```

```
      ID
-----
      100
```

```
SQL> select f.file_name from dba_extents e, dba_data_files f where e.file_id = f.file_id and e.segment_name =
'EMP7' and e.owner = 'HR';
```

```
FILE_NAME
-----
```

```
/u01/app/oracle/oradata/ORCL/users01.dbf
```

장애 발생

```
SQL> ! ls /u01/app/oracle/oradata/ORCL/users01.dbf
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/users01.dbf: No such file or directory
```

```
SQL> ! ls /u01/app/oracle/oradata/ORCL/users01.dbf
```

```
/u01/app/oracle/oradata/ORCL/users01.dbf
```

```
SQL> ! rm /u01/app/oracle/oradata/ORCL/users01.dbf
```

```
SQL> ! ls /u01/app/oracle/oradata/ORCL/users01.dbf
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/users01.dbf: No such file or directory
```

```
SQL> create table hr.emp8 tablespace users as select * from hr.employees;
```

```
create table hr.emp8 tablespace users as select * from hr.employees
```

*

```
ERROR at line 1:
```

```
ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

```
ORA-01116: error in opening database file 7
```

```
ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

```
ORA-27041: unable to open file
```

```
Linux-x86_64 Error: 2: No such file or directory
```

```
Additional information: 3
```

```
SQL> set linesize 150
```

```
SQL> col tbs_name for a15
```

```
SQL> col file_name for a60
```

```
SQL> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

```
SQL> /
```

FILE#	TBS_NAME	FILE_NAME
CHECKPOINT_CHANGE#	STATUS	

1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf
2428588	SYSTEM	
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
2428588	ONLINE	
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
2428588	ONLINE	
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf
2428588	ONLINE	

```
SQL> alter tablespace users offline immediate ;
```

normal : 사용하지 못하게 체크포인트를 발생시킴(patial checkpoint 발생)

temporary : 가능하면 체크포인트 발생시키고 아니면 말고

immediate : 즉시 오프라인

```
SQL> alter tablespace users offline immediate ;
```

ERROR:

ORA-03114: not connected to ORACLE

```
alter tablespace users offline immediate
```

*

ERROR at line 1:

ORA-03135: connection lost contact

Process ID: 2260

Session ID: 17 Serial number: 55017

```
SQL> conn / as sysdba
```

Connected to an idle instance.

```
SQL> startup
```

ORACLE instance started.

Total System Global Area 1543500144 bytes
 Fixed Size 8896880 bytes
 Variable Size 905969664 bytes
 Database Buffers 620756992 bytes
 Redo Buffers 7876608 bytes

Database mounted.

ORA-01157: cannot identify/lock data file 7 - see DBWR trace file

ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/users01.dbf'

SQL> select * from v\$recover_file;

FILE#	ONLINE	ONLINE_	ERROR
CHANGE# TIME	CON_ID		
7	ONLINE	ONLINE	FILE NOT FOUND
0	0		

SQL> alter tablespace users offline immediate;

alter tablespace users offline immediate

*

ERROR at line 1:

ORA-01109: database not open

SQL> alter database datafile 7 offline;

Database altered.

SQL> alter database open;

Database altered.

SQL> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v\$datafile a,
 v\$tablespace b where a.ts# = b.ts#;

FILE#	TBS_NAME	FILE_NAME
CHECKPOINT_CHANGE#	STATUS	
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
3382302	ONLINE	

```

1 SYSTEM /u01/app/oracle/oradata/ORCL/system01.dbf
3382302 SYSTEM
2 UNDOTBS1 /u01/app/oracle/oradata/ORCL/undotbs01.dbf
3382302 ONLINE
7 USERS /u01/app/oracle/oradata/ORCL/users01.dbf
3383221 RECOVER

```

■ 최근 백업파일을 restore

```

SQL> !cp -v /home/oracle/backup/open_bkp/users01.dbf /u01/app/oracle/oradata/ORCL/
'/home/oracle/backup/open_bkp/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'

```

■ 백업 이후 변경정보(리두) 적용

```

SQL> recover tablespace users;
Media recovery complete.

```

- 대상 테이블스페이스 online

```

SQL> alter tablespace users online;

```

Tablespace altered.

- 복구 정보 확인

```

SQL> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;

```

FILE#	TBS_NAME	FILE_NAME
CHECKPOINT_CHANGE#	STATUS	

```

3 SYSAUX /u01/app/oracle/oradata/ORCL/sysaux01.dbf
3382302 ONLINE
1 SYSTEM /u01/app/oracle/oradata/ORCL/system01.dbf
3382302 SYSTEM
2 UNDOTBS1 /u01/app/oracle/oradata/ORCL/undotbs01.dbf
3382302 ONLINE
7 USERS /u01/app/oracle/oradata/ORCL/users01.dbf
3383221 ONLINE

```

```
SQL> select * from hr.emp6;
```

===== 다른 상황 =====

```
SQL> !cp -v /home/oracle/backup/open_bkp/users01.dbf /u01/app/oracle/oradata/ORCL/  
'/home/oracle/backup/open_bkp/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

```
SQL> recover tablespace users;
```

ORA-00279: change 2424496 generated at 05/09/2025 15:24:18 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/0001_0000000013_1200585133.arc

ORA-00280: change 2424496 for thread 1 is in sequence #13

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

auto

ORA-00279: change 2424875 generated at 05/09/2025 15:30:55 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/0001_0000000014_1200585133.arc

ORA-00280: change 2424875 for thread 1 is in sequence #14

ORA-00279: change 2425017 generated at 05/09/2025 15:32:01 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/0001_0000000015_1200585133.arc

ORA-00280: change 2425017 for thread 1 is in sequence #15

ORA-00279: change 2425115 generated at 05/09/2025 15:32:01 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/0001_0000000016_1200585133.arc

ORA-00280: change 2425115 for thread 1 is in sequence #16

ORA-00279: change 2425216 generated at 05/09/2025 15:32:07 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/0001_0000000017_1200585133.arc

ORA-00280: change 2425216 for thread 1 is in sequence #17

ORA-00279: change 2425311 generated at 05/09/2025 15:32:10 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/0001_0000000018_1200585133.arc

ORA-00280: change 2425311 for thread 1 is in sequence #18

Log applied.

Media recovery complete.

```
SQL> alter tablespace users online;
```

Tablespace altered.

```
SQL> select * from hr.emp6;
```

<<시나리오 3>> 오프라인시 테이블스페이스가 손상되었을 경우

■ 장애발생

```
SYS@orcl> shut immediate
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/users01.dbf
/u01/app/oracle/oradata/orcl/users01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/users01.dbf
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/users01.dbf
ls: cannot access /u01/app/oracle/oradata/orcl/users01.dbf: No such file or directory
```

```
SYS@orcl> startup
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes
Fixed Size                  8896880 bytes
Variable Size               905969664 bytes
Database Buffers           620756992 bytes
Redo Buffers                7876608 bytes
```

```
Database mounted.
```

```
ORA-01157: cannot identify/lock data file 7 - see DBWR trace file
```

```
ORA-01110: data file 7: '/u01/app/oracle/oradata/orcl/users01.dbf'
```

```
SYS@orcl> select * from v$recover_file;
```

FILE#	ONLINE	ONLINE_	ERROR
CHANGE# TIME	CON_ID		
7	ONLINE	ONLINE	FILE NOT FOUND
0	0		

1. 문제 발생한 데이터파일 offline

```
SYS@orcl> alter database datafile 7 offline;
```

```
또는
```

```
SYS@orcl> alter database datafile '/u01/app/oracle/oradata/orcl/users01.dbf' offline;
```

```
Database altered.
```



```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,  
v$tablespace b where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME
CHECKPOINT_CHANGE#	STATUS	

1	SYSTEM	/u01/app/oracle/oradata/orcl/system01.dbf
3384241	SYSTEM	
3	SYSAUX	/u01/app/oracle/oradata/orcl/sysaux01.dbf
3384241	ONLINE	
2	UNDOTBS1	/u01/app/oracle/oradata/orcl/undotbs01.dbf
3384241	ONLINE	
7	USERS	/u01/app/oracle/oradata/orcl/users01.dbf
3384241	OFFLINE	

```
SYS@orcl> alter database open;
```

Database altered.

■ 복구 작업

2. 최근 백업 파일을 restore

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/users01.dbf /u01/app/oracle/oradata/orcl/  
'/home/oracle/backup/open_bkp/users01.dbf' -> '/u01/app/oracle/oradata/orcl/users01.dbf'
```

3. 백업 이후 변경정보(리두) 적용

```
SYS@orcl> recover datafile 7;
```

Media recovery complete.

또는

```
SYS@orcl> recover datafile '/u01/app/oracle/oradata/orcl/users01.dbf'
```

또는

```
SYS@orcl> recover tablespace users;
```

4. 대상 데이터파일에 대해서 online

```
SYS@orcl> alter database datafile 7 online;
```

Database altered.

또는

```
SYS@orcl> alter database datafile '/u01/app/oracle/oradata/orcl/users01.dbf' online;
```

또는

```
SYS@orcl> alter tablespace users online;
```

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,  
v$tablespace b where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME
CHECKPOINT_CHANGE#	STATUS	

3	SYSAUX	/u01/app/oracle/oradata/orcl/sysaux01.dbf
3384244	ONLINE	
1	SYSTEM	/u01/app/oracle/oradata/orcl/system01.dbf
3384244	SYSTEM	
2	UNDOTBS1	/u01/app/oracle/oradata/orcl/undotbs01.dbf
3384244	ONLINE	
7	USERS	/u01/app/oracle/oradata/orcl/users01.dbf
3385032	ONLINE	

```
SYS@orcl> alter system archive log current;
```

System altered

<<시나리오 4>> 데이터베이스가 정상적인 종료 후 여러 데이터파일이 손상되었을 경우

```
SYS@orcl> update hr.employees set salary=7777 where employee_id=102;
```

1 row updated.

```
SYS@orcl> select salary from hr.employees where employee_id=102;
```

```
      SALARY
-----
      7777
```

```
SYS@orcl> commit;
```

Commit complete.

```
SYS@orcl> select table_name,tablespace_name
2  from dba_tables
3  where owner='HR' and table_name='EMPLOYEES';
```

```
TABLE_NAME
-----
```

```
TABLESPACE_NAME
-----
```

```
EMPLOYEES
SYSaux
```

-- 대상 파일

/u01/app/oracle/oradata/orcl/users01.dbf

/u01/app/oracle/oradata/orcl/sysaux01.dbf

```
SYS@orcl> shutdown immediate
```

Database closed.

Database dismounted.

ORACLE instance shut down.

■ 장애발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/users01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/sysaux01.dbf
```

```
SYS@orcl> startup
```

ORACLE instance started.

Total System Global Area 1543500144 bytes
Fixed Size 8896880 bytes
Variable Size 905969664 bytes
Database Buffers 620756992 bytes
Redo Buffers 7876608 bytes

Database mounted.

ORA-01157: cannot identify/lock data file 3 - see DBWR trace file

ORA-01110: data file 3: '/u01/app/oracle/oradata/orcl/sysaux01.dbf'

-- 문제가 발생한 파일 정보 확인

-- Alert log file

2025-05-10T20:15:49.827972+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_ora_12894.trc:

ORA-01157: cannot identify/lock data file 3 - see DBWR trace file

ORA-01110: data file 3: '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'

2025-05-10T20:15:49.828433+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_ora_12894.trc:

ORA-01157: cannot identify/lock data file 3 - see DBWR trace file

ORA-01110: data file 3: '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'

ORA-1157 signalled during: ALTER DATABASE OPEN...

2025-05-10T20:15:49.866530+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_12896.trc:

ORA-01110: data file 3: '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'

ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

2025-05-10T20:15:50.001560+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_12896.trc:

ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/users01.dbf'

ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/users01.dbf'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

SYS@orcl> select * from v\$recover_file;

FILE#	ONLINE	ONLINE_ ERROR	CHANGE#	TIME	CON_ID
3	ONLINE	ONLINE	FILE NOT FOUND	0	0
7	ONLINE	ONLINE	FILE NOT FOUND	0	0

문제가 발생한 파일 offline

```
SYS@orcl> alter database datafile 3 offline;
```

Database altered.

```
SYS@orcl> alter database datafile 7 offline;
```

Database altered.

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2458850
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2458850
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2458850
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2458850

```
SYS@orcl> alter database open;
```

Database altered.

Recovery

■ restore

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/{sysaux01,users01}.dbf /u01/app/oracle/oradata/ORCL/
'/home/oracle/backup/open_bkp/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
'/home/oracle/backup/open_bkp/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

2. 백업 이후 변경정보(리두) 적용

```
SYS@orcl> recover datafile 3;
```

```
ORA-00279: change 2356080 generated at 05/10/2025 19:36:51 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_31_1200525254.arc
```

```
ORA-00280: change 2356080 for thread 1 is in sequence #31
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
```

```
auto
```

```
Log applied.
```

```
Media recovery complete.
```

```
SYS@orcl> recover datafile 7;
```

```
ORA-00279: change 2356080 generated at 05/10/2025 19:36:51 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_31_1200525254.arc
```

```
ORA-00280: change 2356080 for thread 1 is in sequence #31
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
```

```
auto
```

```
Log applied.
```

```
Media recovery complete.
```

3. 대상 데이터파일 online

```
SYS@orcl> alter database datafile 3 online;
```

```
Database altered.
```

```
SYS@orcl> alter database datafile 7 online;
```

```
Database altered.
```

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,  
v$tablespace b where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2459444
ONLINE			

1 SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2458853
SYSTEM		
4 UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2458853
ONLINE		
7 USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2459453
ONLINE		

- 살아있네!!

```
SYS@orcl> select salary from hr.employees where employee_id=102;
```

```
SALARY
```

```
-----
```

```
7777
```

```
SYS@orcl> alter system archive log current;
```

System altered.

<<시나리오4>> 데이터베이스 정상적인 종료 후에 system datafile 손상되었을 경우

```
SYS@orcl> select salary from hr.employees where employee_id=102;
```

```
      SALARY
-----
      7777
```

```
SYS@orcl> update hr.employees set salary=8888 where employee_id=102;
```

1 row updated.

```
SYS@orcl> select salary from hr.employees where employee_id=102;
```

```
      SALARY
-----
      8888
```

```
SYS@orcl> commit;
```

Commit complete.

```
SYS@orcl> alter system switch logfile;
```

System altered.

```
SYS@orcl> /
```

System altered.

```
SYS@orcl> /
```

System altered.

```
SYS@orcl> /
```

System altered.

- Archive 확인

```
SYS@orcl> !ls -al /home/oracle/arch*
```


/home/oracle/arch1:

SYS@orcl> shutdown immediate

Database closed.

Database dismounted.

ORACLE instance shut down.

■ 장애 발생

SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/system01.dbf

SYS@orcl> startup

ORACLE instance started.

Total System Global Area 1543500144 bytes

Fixed Size 8896880 bytes

Variable Size 905969664 bytes

Database Buffers 620756992 bytes

Redo Buffers 7876608 bytes

Database mounted.

ORA-01157: cannot identify/lock data file 1 - see DBWR trace file

ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'

-- Alert log file

2025-05-10T21:23:07.558243+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_17539.trc:

ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'

ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/system01.dbf'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

Checker run found 1 new persistent data failures

SYS@orcl> select * from v\$recover_file;

FILE#	ONLINE	ONLINE_ ERROR	CHANGE#	TIME	CON_ID
1	ONLINE	ONLINE	FILE NOT FOUND	0	0

```
SYS@orcl> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
MOUNTED
```

■ Recovery

1. restore

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/system01.dbf /u01/app/oracle/oradata/ORCL/  
'/home/oracle/backup/open_bkp/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

2. 백업 이후 변경정보(리두)적용

```
SYS@orcl> recover datafile 1;
```

```
ORA-00279: change 2356080 generated at 05/10/2025 19:36:51 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_31_1200525254.arc
```

```
ORA-00280: change 2356080 for thread 1 is in sequence #31
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
```

```
auto
```

```
ORA-00279: change 2356143 generated at 05/10/2025 19:38:50 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_32_1200525254.arc
```

```
ORA-00280: change 2356143 for thread 1 is in sequence #32
```

```
ORA-00279: change 2456244 generated at 05/10/2025 19:46:43 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_33_1200525254.arc
```

```
ORA-00280: change 2456244 for thread 1 is in sequence #33
```

```
ORA-00279: change 2457117 generated at 05/10/2025 19:49:01 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_34_1200525254.arc
```

```
ORA-00280: change 2457117 for thread 1 is in sequence #34
```

```
ORA-00279: change 2459505 generated at 05/10/2025 20:26:20 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_35_1200525254.arc
```

```
ORA-00280: change 2459505 for thread 1 is in sequence #35
```

```
ORA-00279: change 2463463 generated at 05/10/2025 21:20:30 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_36_1200525254.arc
```

```
ORA-00280: change 2463463 for thread 1 is in sequence #36
```

Log applied.

Media recovery complete.

3. 데이터베이스 오픈

```
SYS@orcl> alter database open;
```

Database altered.

- 복구 정보 확인

```
SYS@orcl> select status from v$instance;
```

STATUS

OPEN

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2463719
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2463719
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2463719
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2463719

- 잘 살아있네!!

```
SYS@orcl> select salary from hr.employees where employee_id=102;
```

SALARY

8888